

DEVA, Romania

WMO: 152300

Lat: 45.8650N

Lon: 22.8989E

Elev: 791

StdP: 14.28

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	15.0	3.8	6.9	13.4	9.2	9.0	17.5	17.6	40.2	14.8	38.6	2.5	180	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	91.2	68.6	87.9	68.7	84.8	67.5	71.8	84.8	70.1	83.1	68.8	81.3	5.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	105.6	76.2	66.1	99.0	74.6	64.6	94.0	73.0	36.0	84.9	34.5	83.4	33.4	81.8	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.1	10.1	2.9	95.4	7.1	4.1	-2.2	98.3	-6.4	100.7	-10.4	103.0	-15.5	105.9		
(5)			2.5	74.8	7.0	2.4	-2.6	76.5	-6.7	77.9	-10.6	79.3	-15.7	81.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	30.5	35.0	43.0	53.2	60.8	67.3	70.3	70.5	61.9	52.1	41.8	33.0		
	DBStd	15.72	8.09	8.62	7.85	6.60	5.97	5.60	5.27	5.33	6.66	7.05	7.98	7.50		
	HDD50	2159	603	420	237	42	2	0	0	0	1	61	266	527		
	HDD65	5391	1068	839	684	354	154	40	15	14	135	401	696	991		
	CDD50	2786	0	1	19	139	338	519	630	634	359	127	20	0		
	CDD65	543	0	0	0	1	25	109	180	183	43	2	0	0		
	CDH74	5631	0	0	0	59	333	1082	1825	1822	481	29	0	0		
	CDH80	2006	0	0	0	9	53	335	710	753	144	2	0	0		
	WSAvg	3.6	3.1	3.5	4.4	4.2	3.9	3.8	3.7	3.4	3.5	3.2	3.2	3.3		
	PrecAvg	24.5	1.2	1.1	1.4	2.1	2.8	3.2	3.1	2.5	2.3	1.8	1.5	1.6		
(15) Precipitation	PrecMax	31.9	2.2	3.4	3.3	4.3	5.8	5.9	5.8	8.5	4.7	4.1	3.2	4.5		
	PrecMin	15.2	0.3	0.2	0.3	0.2	1.2	0.7	0.4	0.4	0.1	0.1	0.0	0.1		
	PrecStd	4.6	0.6	0.7	0.8	1.0	1.1	1.3	1.4	1.7	1.2	1.0	0.7	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	60.1	70.0	80.5	84.6	91.0	95.3	96.4	90.2	77.5	66.0	53.6		
		MCWB	45.6	49.1	52.6	59.6	64.5	70.5	69.3	68.4	65.6	60.7	55.3	48.1		
	2%	DB	46.7	54.4	65.0	74.2	81.1	87.2	91.0	91.5	83.7	72.8	61.0	49.0		
		MCWB	42.8	46.0	50.7	56.7	63.2	69.4	69.0	68.1	64.4	59.6	53.0	45.4		
	5%	DB	43.1	50.3	60.3	70.1	77.9	84.1	87.4	87.8	79.5	68.4	57.2	45.5		
		MCWB	40.6	44.1	48.3	54.6	62.4	68.0	68.8	68.5	63.3	57.4	51.3	43.1		
	10%	DB	40.3	46.7	56.2	66.4	74.3	80.7	83.7	84.2	75.3	64.1	53.3	42.8		
		MCWB	38.1	42.3	46.4	52.9	60.5	66.2	67.7	67.3	61.9	55.7	48.9	41.0		
	0.4%	WB	46.6	50.1	54.6	61.2	67.8	74.3	73.8	74.5	69.4	63.5	56.7	49.1		
		MCDB	50.0	58.1	66.5	76.8	80.1	86.4	85.8	87.0	85.7	73.4	62.7	52.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	43.4	47.2	51.5	58.4	65.3	71.1	71.5	71.5	66.4	60.6	54.1	45.9		
		MCDB	45.9	52.9	62.5	71.8	76.8	83.4	84.8	84.6	78.7	69.3	59.2	48.2		
	5%	WB	40.9	44.9	49.4	56.3	63.5	69.1	70.0	69.6	64.6	58.7	52.0	43.2		
		MCDB	43.1	49.2	58.3	66.8	74.5	81.1	83.2	82.3	75.0	66.6	56.8	45.3		
	10%	WB	38.3	42.6	47.4	54.3	61.7	67.4	68.7	68.2	62.8	56.5	49.3	41.0		
		MCDB	40.2	46.9	55.0	63.5	72.3	78.4	81.1	80.9	72.5	63.0	53.0	42.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	14.5	18.6	21.1	21.0	21.6	23.1	23.5	20.9	19.9	14.1	9.7		
		MCDBR	14.3	21.0	26.5	28.6	27.8	27.3	30.0	30.2	29.0	27.3	20.0	14.0		
		MCWBR	10.3	13.1	13.9	13.0	11.9	11.2	10.9	10.1	11.4	14.0	12.2	10.3		
	5% WB	MCDBR	13.3	19.2	23.3	24.7	24.5	25.0	25.9	25.6	24.3	24.0	18.0	13.3		
		MCWBR	9.9	12.5	12.8	11.6	11.4	10.9	10.4	9.9	10.5	12.8	11.2	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.355	0.401	0.462	0.448	0.439	0.458	0.469	0.426	0.383	0.352	0.325		
	taud		2.326	2.247	2.182	2.041	2.144	2.215	2.168	2.132	2.217	2.329	2.366	2.360		
	Ebn at Noon		241	255	259	252	261	264	257	248	248	243	229	228		
	Edh at Noon		25	32	40	50	47	44	46	45	38	30	24	22		
(44) All-Sky Solar Radiation	RadAvg		405	657	1012	1373	1638	1832	1802	1650	1175	819	480	329		
	RadStd		52	83	143	149	154	170	155	162	126	95	62	43		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.42	N/A	N/A	+2.63	N/A	N/A	N/A	-371	+301	+146
(47) Regional (70 neighbors)		+1.24	N/A	N/A	+1.67	N/A	N/A	-214	-328	+221	+112

Nomenclature: See separate page